

where again the first RHS term is the KL divergence of the approximate from the true posterior, and $\mathcal{L}(\theta, \phi; \mathbf{x})$ is the variational lower bound of the marginal likelihood of datapoint i :

$$\mathcal{L}(\theta, \phi; \mathbf{x}^{(i)}) = \int q_\phi(\mathbf{z}|\mathbf{x}) \left(\log p_\theta(\mathbf{x}^{(i)}|\mathbf{z}) + \log p_\theta(\mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x}) \right) d\mathbf{z} \quad (16)$$

The expectations on the RHS of eqs (14) and (16) can obviously be written as a sum of three separate expectations, of which the second and third component can sometimes be analytically solved, e.g. when both $p_\theta(\mathbf{x})$ and $q_\phi(\mathbf{z}|\mathbf{x})$ are Gaussian. For generality we will here assume that each of these expectations is intractable.

Under certain mild conditions outlined in section (see paper) for chosen approximate posteriors $q_\phi(\theta)$ and $q_\phi(\mathbf{z}|\mathbf{x})$ we can reparameterize conditional samples $\tilde{\mathbf{z}} \sim q_\phi(\mathbf{z}|\mathbf{x})$ as

$$\tilde{\mathbf{z}} = g_\phi(\epsilon, \mathbf{x}) \quad \text{with} \quad \epsilon \sim p(\epsilon) \quad (17)$$

where we choose a prior $p(\epsilon)$ and a function $g_\phi(\epsilon, \mathbf{x})$ such that the following holds:

$$\begin{aligned} \mathcal{L}(\theta, \phi; \mathbf{x}^{(i)}) &= \int q_\phi(\mathbf{z}|\mathbf{x}) \left(\log p_\theta(\mathbf{x}^{(i)}|\mathbf{z}) + \log p_\theta(\mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x}) \right) d\mathbf{z} \\ &= \int p(\epsilon) \left(\log p_\theta(\mathbf{x}^{(i)}|\mathbf{z}) + \log p_\theta(\mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x}) \right) \Big|_{\mathbf{z}=g_\phi(\epsilon, \mathbf{x}^{(i)})} d\epsilon \end{aligned} \quad (18)$$

The same can be done for the approximate posterior $q_\phi(\theta)$:

$$\tilde{\theta} = h_\phi(\zeta) \quad \text{with} \quad \zeta \sim p(\zeta) \quad (19)$$

where we, similarly as above, choose a prior $p(\zeta)$ and a function $h_\phi(\zeta)$ such that the following holds:

$$\begin{aligned} \mathcal{L}(\phi; \mathbf{X}) &= \int q_\phi(\theta) (\log p_\theta(\mathbf{X}) + \log p_\alpha(\theta) - \log q_\phi(\theta)) d\theta \\ &= \int p(\zeta) (\log p_\theta(\mathbf{X}) + \log p_\alpha(\theta) - \log q_\phi(\theta)) \Big|_{\theta=h_\phi(\zeta)} d\zeta \end{aligned} \quad (20)$$

For notational conciseness we introduce a shorthand notation $f_\phi(\mathbf{x}, \mathbf{z}, \theta)$:

$$f_\phi(\mathbf{x}, \mathbf{z}, \theta) = N \cdot (\log p_\theta(\mathbf{x}|\mathbf{z}) + \log p_\theta(\mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})) + \log p_\alpha(\theta) - \log q_\phi(\theta) \quad (21)$$

Using equations (20) and (18), the Monte Carlo estimate of the variational lower bound, given datapoint $\mathbf{x}^{(i)}$, is:

$$\mathcal{L}(\phi; \mathbf{X}) \simeq \frac{1}{L} \sum_{l=1}^L f_\phi(\mathbf{x}^{(l)}, g_\phi(\epsilon^{(l)}, \mathbf{x}^{(l)}), h_\phi(\zeta^{(l)})) \quad (22)$$

where $\epsilon^{(l)} \sim p(\epsilon)$ and $\zeta^{(l)} \sim p(\zeta)$. The estimator only depends on samples from $p(\epsilon)$ and $p(\zeta)$ which are obviously not influenced by ϕ , therefore the estimator can be differentiated w.r.t. ϕ . The resulting stochastic gradients can be used in conjunction with stochastic optimization methods such as SGD or Adagrad [DHS10]. See algorithm 1 for a basic approach to computing stochastic gradients.

F.1 Example

Let the prior over the parameters and latent variables be the centered isotropic Gaussian $p_\alpha(\theta) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$ and $p_\theta(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$. Note that in this case, the prior lacks parameters. Let's also assume that the true posteriors are approximately Gaussian with an approximately diagonal covariance. In this case, we can let the variational approximate posteriors be multivariate Gaussians with a diagonal covariance structure:

$$\begin{aligned} \log q_\phi(\theta) &= \log \mathcal{N}(\theta; \mu_\theta, \sigma_\theta^2 \mathbf{I}) \\ \log q_\phi(\mathbf{z}|\mathbf{x}) &= \log \mathcal{N}(\mathbf{z}; \mu_z, \sigma_z^2 \mathbf{I}) \end{aligned} \quad (23)$$

Algorithm 2 Pseudocode for computing a stochastic gradient using our estimator. See text for meaning of the functions f_ϕ , g_ϕ and h_ϕ .

Require: ϕ (Current value of variational parameters)

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 $\mathbf{g} \leftarrow 0$ 
for  $l$  is 1 to  $L$  do
   $\mathbf{x} \leftarrow$  Random draw from dataset  $\mathbf{X}$ 
   $\epsilon \leftarrow$  Random draw from prior  $p(\epsilon)$ 
   $\zeta \leftarrow$  Random draw from prior  $p(\zeta)$ 
   $\mathbf{g} \leftarrow \mathbf{g} + \frac{1}{L} \nabla_\phi f_\phi(\mathbf{x}, g_\phi(\epsilon, \mathbf{x}), h_\phi(\zeta))$ 
end for
return  $\mathbf{g}$ 

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where $\mu_{\mathbf{z}}$ and $\sigma_{\mathbf{z}}$ are yet unspecified functions of $\mathbf{x}$. Since they are Gaussian, we can parameterize the variational approximate posteriors:

$$\begin{aligned}
 q_\phi(\boldsymbol{\theta}) & \text{ as } \tilde{\boldsymbol{\theta}} = \boldsymbol{\mu}_\theta + \boldsymbol{\sigma}_\theta \odot \boldsymbol{\zeta} & \text{where } \boldsymbol{\zeta} &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \\
 q_\phi(\mathbf{z}|\mathbf{x}) & \text{ as } \tilde{\mathbf{z}} = \boldsymbol{\mu}_{\mathbf{z}} + \boldsymbol{\sigma}_{\mathbf{z}} \odot \boldsymbol{\epsilon} & \text{where } \boldsymbol{\epsilon} &\sim \mathcal{N}(\mathbf{0}, \mathbf{I})
 \end{aligned}$$

With $\odot$ we signify an element-wise product. These can be plugged into the lower bound defined above (eqs (21) and (22)).

In this case it is possible to construct an alternative estimator with a lower variance, since in this model $p_\alpha(\boldsymbol{\theta})$, $p_\theta(\mathbf{z})$, $q_\phi(\boldsymbol{\theta})$ and $q_\phi(\mathbf{z}|\mathbf{x})$ are Gaussian, and therefore four terms of f_ϕ can be solved analytically. The resulting estimator is:

$$\begin{aligned}
 \mathcal{L}(\phi; \mathbf{X}) &\simeq \frac{1}{L} \sum_{l=1}^L N \cdot \left(\frac{1}{2} \sum_{j=1}^J \left(1 + \log((\sigma_{\mathbf{z},j}^{(l)})^2) - (\mu_{\mathbf{z},j}^{(l)})^2 - (\sigma_{\mathbf{z},j}^{(l)})^2 \right) + \log p_\theta(\mathbf{x}^{(i)} \mathbf{z}^{(i)}) \right) \\
 &+ \frac{1}{2} \sum_{j=1}^J \left(1 + \log((\sigma_{\boldsymbol{\theta},j}^{(l)})^2) - (\mu_{\boldsymbol{\theta},j}^{(l)})^2 - (\sigma_{\boldsymbol{\theta},j}^{(l)})^2 \right)
 \end{aligned} \tag{24}$$

$\mu_j^{(i)}$ and $\sigma_j^{(i)}$ simply denote the j -th element of vectors $\boldsymbol{\mu}^{(i)}$ and $\boldsymbol{\sigma}^{(i)}$.